Playing Gridshot with an SO-101 Robotic Arm

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***Abstract*— We investigate whether a SO-101 robotic arm, fitted with a custom mouse-gripping attachment, can physically play Gridshot by translating visual targets into real-time mouse movements and clicks. Our system follows a perception-planning-control pipeline: a webcam identifies target and cursor positions on screen, a motion planner converts these into physical mouse movements, and the arm executes the resulting trajectory. We evaluate the system using Gridshot's built-in metrics (score, accuracy, per-hit latency).**

I. PROBLEM STATEMENT

Gridshot is a timed aim-training task in which players move a cursor and click targets as quickly and accurately as possible. In the AimCrane version, three targets appear at a time, and each successful hit causes a replacement target to appear elsewhere on the grid [1]. Our goal is to determine whether an SO-101 robotic arm can match or outperform members of our team in Gridshot by physically moving and clicking a computer mouse [2]. The main challenge is translating target locations on the screen into accurate mouse movements while maintaining stable contact with the mousepad and reliably pressing the mouse button.

We will compare the robot's performance with our own results across repeated rounds using the same game settings. We will measure score and accuracy to evaluate how well the robot balances speed and precision compared with human players.

II. APPROACH

This project is **application-oriented**, applying computer vision, motion planning, and robotic control techniques to a target-aiming application using the SO-101. Unlike prior approaches using other robotic platforms, this project explores using the SO-101 arm to physically control a mouse and play Gridshot [3].

We currently plan on using the traditional robot manipulation technique: $Perception \rightarrow Planning \rightarrow Control$.

A. Perception

We would like to request an **additional webcam** so that we can monitor both the display and the robot arm. One webcam will capture the game screen for real-time image processing, while the other will be used to plan the motion for the arm to interact with the mouse. In the Gridshot aim-training environment, targets appear as bright purple circles, allowing the use of **computer vision** techniques (like color segmentation, object detection) to identify the coordinates of each visible target. The system will also detect the mouse cursor and determine its position relative to the visible targets.

Based on these positions, the system will select the target closest to the current cursor position and provide the target's location to the motion-planning system, which will command the SO-101 arm to move the cursor toward the target. Once the cursor is sufficiently aligned with the target, the system will issue a mouse-click command to hit the target.

B. Motion Planning

We'll first calibrate the relationship between small physical mouse movements and the resulting cursor movements. Using this calibration and the arm's kinematic model, we will plan a trajectory that brings the cursor towards the selected target while respecting robot joint limits and maintaining mousepad contact [4].

C. Manipulation

The SO-101 must reliably grasp and move the mouse while maintaining contact with the mousepad. Since the standard SO-101 gripper may slip while holding the mouse during fast movements, we currently plan to design and 3D-print a custom attachment for the mouse. The attachment will provide a better grip and control with the mouse.

III. SIMULATION ENVIRONMENT

We intend to use simulation environments minimally, as the nature of simulated mouse usage will not have the same fidelity as real-life experimentation. Most of the development and evaluation will be performed using the real robot arm to ensure that the results reflect the conditions of physical mouse manipulation.

IV. EVALUATION

Our primary evaluation metric will be based on the system's ability to reliably/accurately translate visual target locations into physical mouse movements and clicks.

The performance of the robot can be objectively measured by the metrics provided upon completion of a Gridshot session. This includes:

- Game Score
- Successful Hits
- Misses
- Accuracy
- Per-Hit Latency

We aim to maximize the game score, number of successful hits and accuracy, and minimize the number of misses and per-hit latency.

We seek to compare these values to our own personal scores in playing Gridshot. We would consider our project successful if it is better than we are (Game Score > 5000), with further hopes of being comparable to, or better than, the scores of professional esports players.

REFERENCES

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